

Glare Removal & Thin Cloud Detection

Ignore the sun, catch the cirrus.

Algorithmic Sun Glare Masking

- **HSV filter:** Flags extreme brightness ($V > 250$) with near-zero saturation ($S < 20$).
- **Highlight masking:** Blurs and dilates the sun disc, removing glare from cloud detection and region evaluation.
- **Unbiased results:** Glare pixels are excluded from the sky percentage, so sunlight never lowers the clear-sky score.

Wispy Cirrus Detection

- **Tversky loss:** Weighted ($\beta \approx 0.75$) to penalize missed clouds more than false alarms.
- **Voltage protection:** Faint cirrus still cuts panel voltage, so reducing false negatives captures it.
- **Circular ROI:** Percentages are computed only inside an adjustable green aperture matching the solar collection angle.